

Roll No.:

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech

Semester: III

Name of the Course: Fundamentals of Information Security and Block Chain

Course Code: TCS 332

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Explain various methods to protect against unauthorized modification, deletion, and access.	CO1
(b)	Describe the phases involved in penetration testing and explain how each phase contributes to system security.	CO1
(c)	Compare white hat, grey hat, and black hat hackers in terms of their intent, approach, and legality.	CO1
Q2		
(a)	Write and explain Linux commands for creating, copying, moving, and deleting files and directories.	CO2
(b)	Write a program to check if an entered password is strong (length, digits, special characters).	CO2
(c)	Write a Bash Script to sort an integer array in ascending order.	CO2
Q3		
(a)	Explain the TCP 3-way handshake with a neat diagram and example packet exchange.	CO4
(b)	Explain Phishing and Spear Phishing with examples. How can organizations mitigate them?	CO3
(c)	Write a short case study analyzing how combined phishing and SQL injection attacks compromise systems.	CO3
Q4		
(a)	Write a Python program to create a simple blockchain with 3 blocks using SHA-256 for hashing.	CO5
(b)	Explain the structure of a block with all its fields (block header, nonce, timestamp, Merkle root).	CO5
(c)	What is the genesis block? Explain its importance in blockchain history.	CO4
Q5		
(a)	Compare and contrast Proof-of-Work (PoW) and Proof-of-Stake (PoS) with suitable examples.	CO5
(b)	What is a Distributed Ledger? Explain its benefits over centralized systems.	CO5
(c)	Describe the concept of Smart Contracts with an example from Ethereum.	CO6

